Autoresearch findings

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Abstract

This report collects the findings: An unexploitable modal agent that defects against itself; When graded PrudentBots cooperate with graded FairBots.

1 An unexploitable modal agent that defects against itself

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent in Peano arithmetic, written with the provability box $\Box$ of the logic GL. FairBot, $\Box p$, cooperates exactly when it can prove that the opponent cooperates with it (p); CooperateBot always cooperates; DefectBot always defects. An agent is *unexploitable* if no modal opponent ever gets it to cooperate while the opponent defects. We consider agents built from p, from q ("the opponent cooperates with DefectBot"), $\bot$, $\neg$, $\wedge$, $\vee$ and $\Box$, with every atom under a box, and measure size as the number of nodes in the formula.

Result

The agent $W = \Box(\Box\Box\bot \vee (p \wedge (q \vee \neg\Box q)))$ is unexploitable, cooperates with FairBot and with CooperateBot, yet defects against an exact copy of itself. An exhaustive search shows that no agent of size 11 or less has this combination of properties, and W (size 12) is one of the three smallest. It is the first member of an infinite family $W_n = \Box(\Box^n\bot \vee (p \wedge (q \vee \neg\Box q)))$, $n \geq 2$, all with the same four properties, with exact laws for how W_n plays against graded FairBots $\Box^k p$, graded PrudentBots and other members of the family.

Why it matters

Every unexploitable agent in the literature we checked (FairBot, PrudentBot, CliqueBot) cooperates with its own copy. W_n shows that unexploitability, cooperation with FairBot and generosity towards CooperateBot do not force self-cooperation: the delayed "gate" $\Box^n\bot$ lets the agent tolerate FairBot's failure

to prove anything about DefectBot, while two copies of W_n never reach mutual cooperation.

Proved vs checked

The properties of W_n and its laws against graded FairBots, graded PrudentBots and other W_m are proved for all indices in Lean 4 (height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [6], cited, not formalised; see also [2]). The minimality of size 12 is an exhaustive computation over all agents up to size 12 in the stated language.

We believe this result is new. Our prior-art search (2026-10-17) covered:

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post "Robust Cooperation in the Prisoner's Dilemma" and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling" (Alignment Forum).
- Mennen, "Modal Combat for games other than the prisoner's dilemma" (agentfoundations.org item 1304).
- github.com/machine-intelligence/provability (Haskell solver and combat code; the agents folder contents were not visible).

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the proof and check files are included so anyone can re-run them.

Claim K6

In L (atoms p,q boxed; F,~,&,|,[]; size=nodes), the smallest agent that cooperates with FairBot and CooperateBot, defects against itself and is unexploitable has size 12; W=[]([][]F|(p&(q|~[]q))) is one, unexploitable by every opponent; none of size <=11.

Status: derived.

Verified.

- `checks/K6-c046_combo12.py`: census counts to size 12; S=0,FBc=1,CBc=1 counts per size 4..12 = 2,2,4,8,19,46,120,357,900; none <=11 and exactly 3 of size 12 unexploited by all 2815 size<=9 classes; W's self/FB/CB/DB outcomes and zero exploiters by an independent height-table solver; GLS certificates for W vs W, FB, CB with flipped-candidate controls — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K9

For every n>=2, the agent W_n=[](□^n F|(p&(q|~[]q))) of size n+10 is unexploitable by every modal opponent, mutually cooperates with FairBot and CooperateBot, and defects against itself (both copies defect above height n+1); W_0, W_1 fail with FairBot. Lean-proved.

Status: derived.

Verified.

- `checks/K9-c046_wn_lean.py`: embedded Lean 4 theorems w_gate, w_unexploitable, w_db, w_cb, w_self, w_self_db, w_fb compile with no sorryAx; three false variants rejected; height tables n<=12 and 3000 random opponent traces agree; size(W_n)=n+10 — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K11

For all n>=0, k>=1, j,m>=0 (height semantics of closed GL fixed points): W_n and F_k cooperate at every height iff k<n; W_n and P_j iff 2<=n<j; W_n and W_m both defect from height n+m+3. Hence all W_n (n>=2) are distinct. Lean-proved.

Status: derived.

Verified.

- `checks/K11-c046_wpairs_lean.py`: embedded Lean 4 theorems w_fk, pj_db, w_pj, w_wm compile with no sorryAx; four false variants rejected with proof errors; height tables n<=12, k,j,m<=14 (heights 0..79) agree with all theorems and K10's n=2..5 predicates — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261017-1126`, claims K6, K9, K11, run-folder commit `662166ecff0651902cc5fcb04bf747c62d795af2`, published 2026-10-18.

2 When graded PrudentBots cooperate with graded FairBots

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. A *modal agent* [1] decides by asking what Peano arithmetic (PA) can prove about the opponent, written with the provability box $\Box$ of the logic GL. Here p means "the opponent cooperates with me" and q means "the opponent cooperates with DefectBot", the agent that always defects. FairBot, $\Box p$, cooperates exactly when it can prove that the opponent cooperates with it. PrudentBot adds a safety test: it also requires a proof, in the stronger system PA+1, that the opponent would not cooperate with DefectBot, so it does not reward an opponent that cooperates with everyone.

We study two graded families. The graded FairBot $F_k = \Box^k p$ $(k \geq 1)$ asks for k nested layers of provability; F_1 is FairBot. The graded PrudentBot $P_j = \Box p \wedge \Box(\neg\Box^j \bot \to \neg q)$ $(j \geq 0)$ runs its safety test in PA+j, since $\neg\Box^j\bot$ expresses the consistency of PA+$(j-1)$; P_1 is PrudentBot.

Result

P_j and F_k cooperate with each other exactly when $k \leq j$; when $k > j$, both defect. In words: a PrudentBot that runs its safety test in a stronger system can afford to cooperate with FairBots that demand more nested proof, and the exchange rate is one for one. Every P_j still defects against CooperateBot, so the extra cooperation costs nothing in safety.

Why it matters

Barasz et al. remark that PrudentBot's safety test may use a stronger system than PA+1 and that this widens the set of agents it cooperates with, and online discussion of the paper looked at specific pairs of such variants. Here the relation is pinned down exactly for the two graded families, for all indices: the boundary of cooperation is the single inequality $k \leq j$.

Proved vs checked

Proved for all $j \geq 0$ and $k \geq 1$ in Lean 4, in the height semantics of closed GL fixed points (on the chain of heights, both agents cooperate at every height when $k \leq j$; when $k > j$ both defect above height $j + k$). The link to provability in PA is Solovay's theorem [6] (see also [2]), cited, not formalised, and the reading in the standard model assumes PA is sound.

This result extends # Prior-art answer for R4 (K8: P_j / F_k threshold law)

Verdict: partial prior art.

- Barasz et al. 2014, remark after Theorem 3.2: PrudentBot's DefectBot check may use PA+n for larger n, which widens the set of agents it cooperates with without losing safety. That is the idea behind graded PrudentBots, stated informally.
- LessWrong comments on the Barasz et al. post (orthonormal, cousin_it) discuss indexed variants "Prudent(m,n)Bot" with specific pairs, e.g. Prudent(0,100)Bot cooperating with Prudent(0,1)Bot and Prudent(99,1000)Bot, and cousin_it noting that a PrudentBot using PA and PA+1 does not cooperate with one using PA+1 and PA+2.
- Not found: a general threshold law for P_j against graded FairBots F_k (cooperate iff k <= j, both defect above j+k), or a proof of it.
- Suggested wording: the graded family was discussed informally (Barasz et al. remark; LessWrong comments); the general law and its Lean proof were not found.

Sources searched (owner, web search, 2026-10-17)

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post "Robust Cooperation in the Prisoner's Dilemma" and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
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- Mennen, "Modal Combat for games other than the prisoner's dilemma" (agentfoundations.org item 1304).
- github.com/machine-intelligence/provability (Haskell solver and combat code; the agents folder contents were not visible).

Not searched: Critch's parametric bounded Löb paper, theses, the contents of the Haskell agents folder, non-English sources. A "none" verdict means "not found in this search", nothing more. Our prior-art search (2026-10-17) covered:

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.

- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post "Robust Cooperation in the Prisoner's Dilemma" and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
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- github.com/machine-intelligence/provability (Haskell solver and combat code; the agents folder contents were not visible).

If you know of related work we missed, please open an issue.

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Claim K8

For all j>=0, k>=1 (height-chain semantics of closed GL fixed points): P_j=[]p&[](~[]^jF->~q) and F_k=[]^k p cooperate at every height iff k<=j; if j<k both defect above height j+k. Lean-proved; P_j defects vs CB.

Status: derived.

Verified.

- `checks/K8-c046_pjfk_lean.py`: embedded Lean 4 proof of pj_fk and pj_cb compiles with no sorryAx; flipped-conclusion variant is rejected by Lean; explicit height table for j<=8,k<=8 matches (coop everywhere iff k<=j, else both defect above j+k with exact threshold) — pass (Lean-proved, no sorryAx)

Prior art. Partial prior art.

Provenance. Run `20261017-1126`, claims K8, run-folder commit `662166ecff0651902cc5fcb04bf747c62d79` published 2026-10-18.

3 How to verify

Each finding has its check scripts in `claims/<slug>/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[3] Andrew Critch. A parametric, resource-bounded generalization of löb's theorem, and a robust cooperation criterion for open-source game theory. *Journal of Symbolic Logic*, 2019.

[4] Patrick LaVictoire, Benja Fallenstein, Eliezer Yudkowsky, Mihaly Barasz, Paul Christiano, and Marcello Herreshoff. Program equilibrium in the prisoner's dilemma via löb's theorem. In *AAAI Workshop on Multiagent Interaction without Prior Coordination*, 2014.

[5] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[6] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.